

Solutions for Calculus Vol 1: One variable
calculus, with an introduction to Linear Algebra
(2nd Edition) by Tom M. Apostol

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0.1 Introduction

0.1.1 1.4 Exercises

1

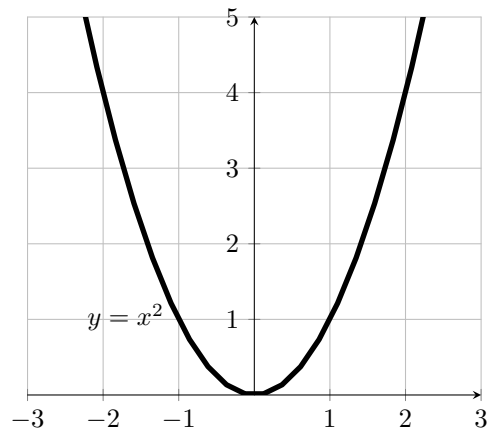
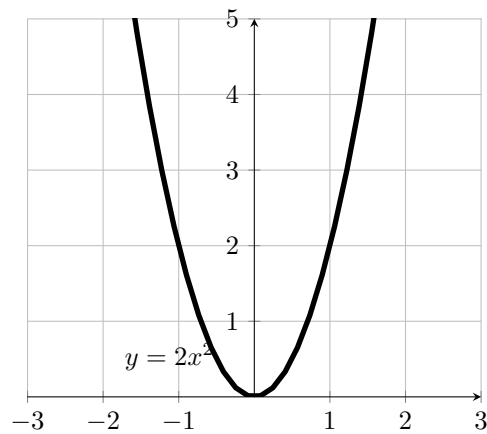


Figure 1.3: $y = x^2$

a) Modify the region in Figure 1.3 by assuming that the ordinate at each x is $2x^2$ instead of x^2 .

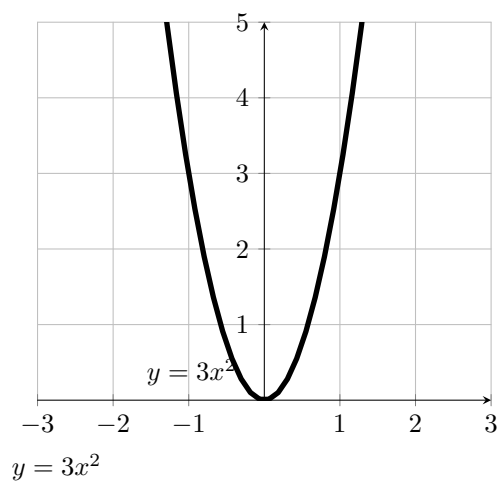
Draw the new figure.



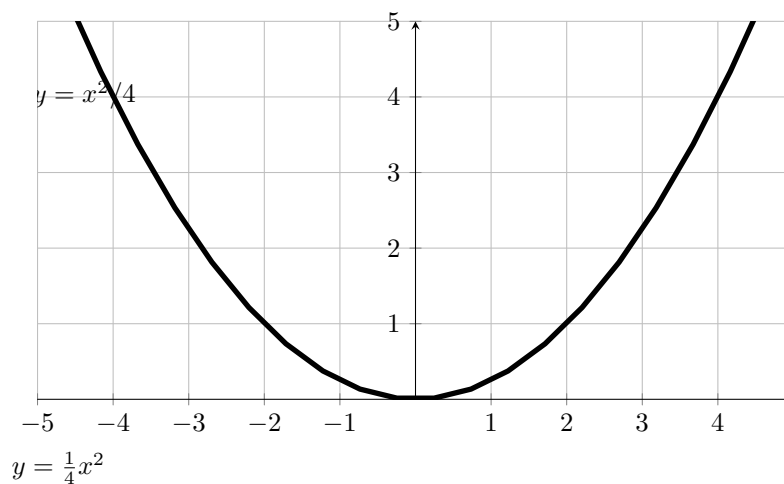
$y = 2x^2$

Check through the principal steps in the forgoing section and find what effect this has on the calculation of the area. Do the same if the ordinate at each x

b) $3x^2$



c) $\frac{1}{4}x^2$



d) $2x^2 + 1$

e) $ax^2 + c$

2

Modify the region in Figure 1.3 by assuming that the ordinate at each x is x^3 instead of x^2 .

Draw the new figure.

a) Use a construction similar to that illustrated in Figure 1.5 and show that the outer and inner sums S_n and s_n are given by

$$S_n = \frac{b^4}{n^4}(1^3 + 2^3 + \dots + n^3), \quad s_n = \frac{b^4}{n^4}[1^3 + 2^3 + \dots + (n-1)^3]$$

b)

Chapter 1

The concepts of integral calculus

1.5 Exercises

1.5.1 1

Let $f(x) = x + 1$ for all real x . Compute the following

$$f(2) = 3$$

$$f(-2) = -1$$

$$-f(2) = -3$$

$$f\left(\frac{1}{2}\right) = \frac{3}{2}$$

$$1/f(2) = \frac{1}{3}$$

$$f(a + b) = a + b + 1$$

$$f(a) + f(b) = a + b + 2$$

$$f(a)f(b) = (a + 1)(b + 1) = ab + a + b + 1$$

1.5.2 2

Let $f(x) = 1 + x$ and let $g(x) = 1 - x$ for all real x . Compute the following:

$$f(2) + g(2) = 3 + (-1) = 2$$

$$f(2) - g(2) = 3 - (-1) = 4$$

$$f(2)g(2) = -3$$

$$f(2)/g(2) = -3$$

$$f[g(2)] = f(-1) = 0$$

$$\begin{aligned}
g[f(2)] &= g[3] = -2 \\
f(a) + g(-a) &= 1 + a + (1 - (-a)) = 2 + 2a \\
f(t)g(-t) &= (1 + t)(1 - (-t)) = 1 + 2t + t^2
\end{aligned}$$

1.5.3 3

Let $\psi(x) = |x - 3| + |x - 1|$ for all real x . Compute the following:

$$\begin{aligned}
\psi(0) &= |0 - 3| + |0 - 1| = 3 + 1 = 4 \\
\psi(1) &= |1 - 3| + |1 - 1| = 2 \\
\psi(2) &= |2 - 3| + |2 - 1| = 1 + 1 = 2 \\
\psi(3) &= |3 - 3| + |3 - 1| = 2 \\
\psi(-1) &= |-1 - 3| + |-1 - 1| = 6 \\
\psi(-2) &= |-2 - 3| + |-2 - 1| = 5 + 3 = 8
\end{aligned}$$

Find all t for which $\psi(t + 2) = \psi(t)$ $\psi(t) = |t - 3| + |t - 1|$ $\psi(t + 2) = |t + 2 - 3| + |t + 2 - 1| = |t - 1| + |t + 1|$

$$\begin{aligned}
\text{Given } |t - 3| + |t - 1| &= |t - 1| + |t + 1| \\
\text{thus } |t - 3| &= |t + 1|
\end{aligned}$$

Only value that will satisfy is when $t = 1$

1.5.4 4

Let $f(x) = x^2$ for all real x . Verify each of the following formulas. In each case describe the set of real x , y , t , etc., for which the given formula is valid

- (a) $f(-x) = f(x)$, So $f(-x) = (-x) \cdot (-x) = (x) \cdot (x) = f(x)$ for all x
- (b) $f(y) - f(x) = (y - x)(y + x)$, So $(y - x)(y + x) = y^2 - x^2 = f(y) - f(x)$ for all x and y
- (c) $f(x + h) - f(x) = 2xh + h^2$, So, $f(x + h) - f(x) = (x + h)^2 - x^2 = x^2 + 2xh + h^2 - x^2$ for all x and h
- (d) $f(2y) = 4f(y)$, So $f(2y) = (2y)^2 = 4y^2 = 4f(y)$ for all y
- (e) $f(t^2) = f(t)^2$, So $f(t^2) = (t^2)^2 = f(t)^2$ for all t
- (f) $\sqrt{f(a)} = |a|$, So $\sqrt{f(a)} = \sqrt{a^2}$ which when taking the positive root, is $|a|$ for all a

1.15 Exercises

1.15.1 1

Compute the value of the following integrals

- (a) $\int_{-1}^3 [x] dx = \int_{-1}^0 (-1) dx + \int_0^1 (0) dx + \int_1^2 (1) dx + \int_2^3 (2) dx = -1 + 0 + 1 + 2 = 2$
See Figure 1.1

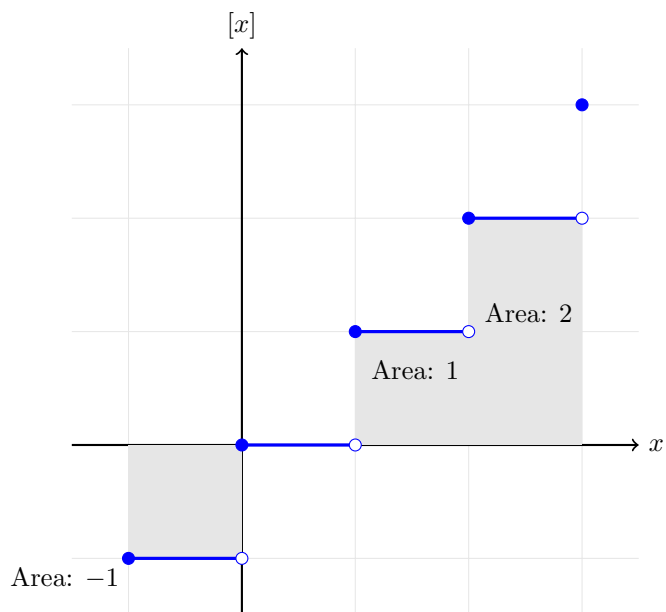


Figure 1.1: Plot of $[x]$ function

1.26 Exercises

Compute the value of the following integrals

1. $\int_0^3 x^2 dx = \left. \frac{x^3}{3} \right|_0^3 = \frac{3^3}{3} - \frac{0^3}{3} = 9$
2. $\int_{-3}^3 x^2 dx = \left. \frac{x^3}{3} \right|_{-3}^3 = \frac{3^3}{3} - \frac{-3^3}{3} = 9 - (-9) = 18$
3. $\int_0^2 4x^3 dx = \left. \frac{4x^4}{4} \right|_0^2 = x^4 \Big|_0^2 = 2^4 - 0 = 16$
4. $\int_{-2}^2 4x^3 dx = x^4 \Big|_{-2}^2 = 2^4 - (-2)^4 = 0$
5. $\int_0^1 5t^4 dt = \left. \frac{5t^5}{5} \right|_0^1 = 1^5 - 0 = 1$
6. $\int_{-1}^1 5t^5 dt = \left. \frac{5t^6}{6} \right|_{-1}^1 = \frac{5}{6} - \frac{5}{6} = 0$
7. $\int_0^1 (5x^4 - 4x^3) dx = \int_0^1 5x^4 dx - \int_0^1 4x^3 dx = x^5 - x^4 \Big|_0^1 = (1^5 - 1^4) - 0 = 0$
8. $\int_{-1}^1 (5x^4 - 4x^3) dx = \int_{-1}^1 5x^4 dx - \int_{-1}^1 4x^3 dx = x^5 - x^4 \Big|_{-1}^1 = (1^5 - 1^4) - (-1^5 - (-1)^4) = 0 - (-1 - 1) = 2$

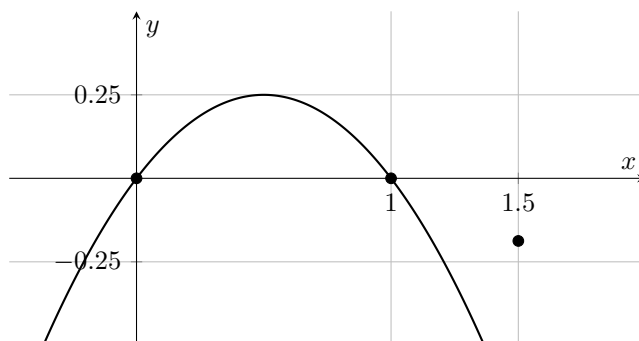


Figure 1.2: Plot of $x(x-1)$ function

$$9. \int_{-1}^2 (t^2 + 1) dt = \left. \frac{t^3}{3} + t \right|_{-1}^2 = \left(\frac{2^3}{3} + 2 \right) - \left(\frac{-1}{3} - 1 \right) = \left(\frac{14}{3} \right) - \left(\frac{-4}{3} \right) = \frac{18}{3} = 6$$

21. Find all values of c for which

$$(a) \int_0^c x(1-x) dx = 0$$

$$\textbf{Solution} \quad \int_0^c x(1-x) dx = \int_0^c (x - x^2) dx = \left. \frac{x^2}{2} - \frac{x^3}{3} \right|_0^c = \left(\frac{c^2}{2} - \frac{c^3}{3} \right) - 0$$

$$\frac{c^2}{2} - \frac{c^3}{3} = 0 = c^2 \left(\frac{1}{2} - \frac{c}{3} \right) = c^2 \left(\frac{3}{6} - \frac{2c}{6} \right) \text{ thus } c = 0 \text{ or } c = 1.5 \text{ Figure??}$$

$$(b) \int_0^c |x(1-x)| dx = 0$$

Solution $c = 0$ as this never has area under the y-axis. See Figure ??